



ANJUMAN COLLEGE OF ENGINEERING TECHNOLOGY

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ABSTRACT: Solar vitality is the foremost abundant, renewable vitality source within the world. As solar energy adoption increases, maintaining the cleanliness of solar panels is crucial for maximizing energy output. The solar panels in case of residential use are usually placed on the roof or terrace to receive the maximum amount of sunlight. Because of this, cleaning these solar panels would require the homeowner climbing up onto the roof to clean the panels, which can be hazardous. The other choice would be to enlist a company to do it for them. In either of the cases it is hazardous and can prove fatal. Thus, the proposed work introduces a design and development of an AI-powered drone system for efficient and autonomous cleaning of solar panels. The main object of this design prototype is to clean the solar panel using an drone system, such that efficiency and quality of solar panel is not compromised. The proposed work will try to introduce a drone base system in which the drone installed with a small water tank would be used to locate the place using mission planner, where the solar panels have been installed and then using the cleaning system installed on the drone to clean the panels.

INTRODUCTION: Solar panels provide sustainable energy as a clean, inexhaustible, and dependable power source. This form of energy can be harnessed for electricity generation in various sectors, including residential, commercial, and industrial applications. Unlike other energy sources, solar power produces no harmful emissions and offers a potential solution for global energy needs without relying on fossil fuels. According to the US Department of Energy, the sunlight that reaches Earth in just 90 minutes could power the entire world for a year. While fossil fuels risk depleting Earth's resources, solar energy does not face this limitation. Solar panels, composed of semiconductors, convert sunlight into direct current electricity. Their pollution-free, renewable, and safe nature has led to increased adoption in recent years. Large-scale solar power facilities have been established in numerous countries, such as Australia, the Middle East, the United States, Europe, and China. However, certain on-site challenges must be addressed, including bird droppings, dust accumulation, water stains, and efficiency reductions due to wiring issues, module soiling, and inverter losses.

The accumulation of dust on solar panels is affected by numerous elements, including the dust particles' biological, electrostatic, and chemical characteristics, as well as their form, dimensions, and mass. Currently, a range of cleaning techniques, such as electrostatic and water-based methods, are commonly employed to remove dust from solar panels.

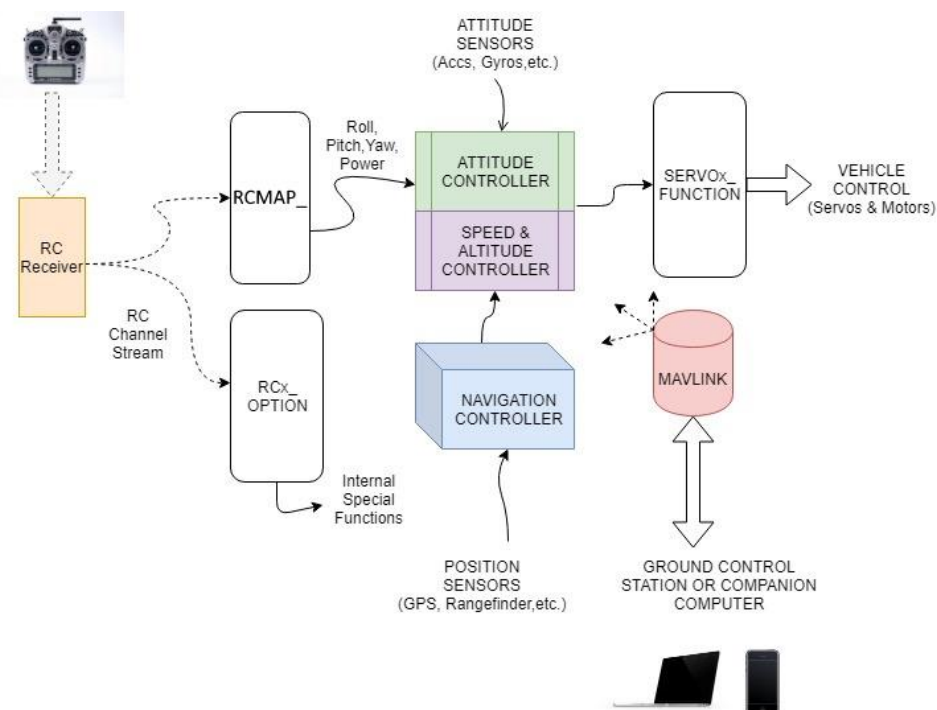
In the mechanism proposed in our paper, we introduce a method which integrates the use of Unmanned Aerial Vehicle and Digital Image Processing (DIP) to clean the panel with the cleaning system installed on the top of the drone.

The objective of this study is to develop and implement a prototype of an autonomous drone equipped with a water reservoir and integrated with Mission Planner software. This system aims to improve the efficiency of solar panels installed in elevated locations, such as rooftops of residential and commercial structures.

HARDWARE IMPEMETATION:



BLOCK DIAGRAM:



CONCLUSION: This study highlights the potential of AI-powered drones as an inventive and independent arrangement for solar panel maintenance. By leveraging automated route and optimized cleaning mechanisms, the prototype ensures more secure, speedier, and more effective cleaning compared to traditional strategies. The capacity to reduce water utilization and operational dangers underscores its practical importance.

Beyond residential and industrial applications, this innovation might be extended to large-scale solar farms, where manual cleaning is labor-intensive and inefficient. Future research could explore integrating AI-driven dirt detection, self-charging drone systems, and versatile cleaning strategies to advance improve performance. The adoption of such systems might contribute to progressing the longevity and productivity of solar panels, reinforcing the part of automation in maintainable energy solutions.

.WORKING: The working of the AI-powered drone for solar panel cleaning involves the integration of flight control, image processing, and an automated spraying mechanism. Initially, the drone is powered on and configured through the Pixhawk flight controller, GPS module, and Mission Planner software, which helps in setting up the flight path and navigation parameters. A camera module connected to the Raspberry Pi 4 captures real-time images of the solar panels. These images are processed using Python-based digital image processing techniques to detect dust or dirt on the panels. Based on the detection and the GPS coordinates, the drone follows a predefined autonomous flight path over the panels. When the drone reaches a dirty section, the spraying system is activated using a VRA switch on the RC transmitter. This system consists of a BLDC motor, a 30A ESC, a water container, and a small pipe that sprays water onto the panel surface to remove dust. The entire operation is monitored in real-time using a telemetry system, which provides updates on the drone's position, battery status, and system health. After completing the cleaning process, the drone safely returns to its base location and lands. This automated workflow ensures efficient, contactless, and smart cleaning of solar panels using drone technology.

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